

## 1. What is the Bias-Variance Tradeoff?

The Bias-Variance Tradeoff is the balance between:

- **High Bias** → Underfitting
- **High Variance** → Overfitting

A good model achieves the right balance for better accuracy.

**Goal:** Find the sweet spot — low bias AND low variance 

## 2. What is the difference between a loss function and a cost function?

| Loss Function                 | Cost Function                         |
|-------------------------------|---------------------------------------|
| Error for a single data point | Average error over the entire dataset |
| Used for one prediction       | Used during model training            |

# Loss — one

$$\text{sample loss} = (y_{\text{actual}} - y_{\text{predicted}})^2$$

# Cost — all

$$\text{samples cost} = (1/n) \times \sum (y_{\text{actual}} - y_{\text{predicted}})^2$$

## 3. What is Gradient Descent?

Gradient Descent is an optimization algorithm that minimizes the loss function by updating model parameters step by step toward the minimum error.

1. Start with random weights
2. Calculate cost
3. Find slope (gradient)
4. Move in opposite direction of slope
5. Repeat until cost is minimum

$$\text{weight} = \text{weight} - (\text{learning\_rate} \times \text{gradient})$$

| Term          | Meaning               |
|---------------|-----------------------|
| Learning rate | Step size             |
| Too high LR   | Overshoots minimum    |
| Too low LR    | Very slow to converge |

## 4. What is a Confusion Matrix?

A Confusion Matrix is a table used to evaluate a classification model's performance.

It contains:

|        |          | Predicted |          |
|--------|----------|-----------|----------|
|        |          | Positive  | Negative |
| Actual | Positive | TP        | FN       |
|        | Negative | FP        | TN       |

- **TP** – True Positive
- **TN** – True Negative
- **FP** – False Positive
- **FN** – False Negative

# Key metrics from confusion matrix

Accuracy =  $(TP + TN) / \text{Total}$

Precision =  $TP / (TP + FP)$

Recall =  $TP / (TP + FN)$

## 5. What are L1 (Lasso) and L2 (Ridge) Regularization?

Techniques to **prevent overfitting** by penalizing large weights.

- **L1 (Lasso):** Adds the absolute value of coefficients as a penalty and can reduce some coefficients to zero (feature selection).
- **L2 (Ridge):** Adds the squared value of coefficients as a penalty and reduces coefficient values without making them zero.

```
from sklearn.linear_model import Lasso, Ridge
```

```
Lasso(alpha=0.1) # L1 — removes useless features
```

```
Ridge(alpha=0.1) # L2 — shrinks all weights
```

“Regularization adds a penalty for complexity — L1 removes features, L2 shrinks them.”

## 6. What is the "Curse of Dimensionality"?

The Curse of Dimensionality refers to problems that arise when the number of features becomes very large, making models slower, more complex, and less accurate.

Model overfits easily

**Fix:**

- Feature selection
- PCA (dimensionality reduction)
- Remove irrelevant columns

## 7. How does Logistic Regression differ from Linear Regression?

| Linear Regression            | Logistic Regression              |
|------------------------------|----------------------------------|
| Predicts continuous values   | Predicts categories/classes      |
| Used for regression problems | Used for classification problems |
| Output is any real value     | Output is a probability (0–1)    |
| Predict salary               | Spam or not spam                 |

# Linear — predicts a number

$y = mx + c$  # e.g. salary = 50000

# Logistic — predicts probability

$y = \text{sigmoid}(mx + c)$

# e.g. spam = 0.87 → spam 

## 8. What is the difference between Parameters and Hyperparameters?

| Parameters              | Hyperparameters                            |
|-------------------------|--------------------------------------------|
| Learned during training | Set before training                        |
| Example: Weights, Bias  | Example: Learning rate, Epochs, Batch size |

## 9. What is the difference between feature engineering and feature selection?

- **Feature Engineering:** Creating or transforming new features from existing data.
- **Feature Selection:** Choosing the most important features while removing irrelevant ones.

## 10. What are the key assumptions of Linear Regression?

The main assumptions are:

- **Linear relationship** between features and target
- **Independence** of observations
- **Constant variance (Homoscedasticity)** of errors
- **Normally distributed residuals**
- **No multicollinearity** among independent variables

“Linear Regression assumes a straight-line relationship, independent features, and normally distributed errors — violating these reduces model reliability.”